Lever Test · TA Brief

Learning goal: I can test how the distance from the pivot changes the effort needed to lift a load.

Watch (staff preview first)

Short silent teacher demonstration clip for this lesson's model and practical sequence · preview before class use.

Adult Support Routine

No cold calling; nobody is asked to read difficult text aloud.

Give time to start -> point to the relevant place -> rephrase -> offer a small prompt or word bank -> reduce to a limited choice.

Beside the You do in the pupil book, mark I (independent), S (supported) or C (choice).

Must-Know Words

effort distance: the distance from the pivot to where the effort push is applied

load: the weight or mass being lifted by the lever

independent variable: the one thing we change on purpose (the effort distance from the pivot)

pattern: how one measurement changes when we change another

Key Answers

A. Doubling the effort distance from the pivot halves

the effort needed to balance and lift the load.

A. Using the pattern, 15 cm lies between 10 cm and 20 cm, so the effort needed is between 4 coins and 2 coins (about 3 coins).

1. 20 cm (needed 2 coins compared with 8 coins at 5 cm). 2. The effort needed decreases. 3. Same load mass, same load distance from the pivot, same ruler/pivot, and identical coins.